A UNIQUENESS CONDITION FOR THE INVERSE BLIND PROBLEM AND THE SYNCOPATED SAMPLING

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Abstract. Magnetic Resonance Imaging (MRI) has gained significant attention in various fields such as medical imaging, materials science, and chemistry. An MRI machine reads radio frequency data across time-consuming scans and then synthesizes the total information to construct the MR image. This process makes MRI expensive and slow. Recent studies have investigated accurate reconstruction methods using a reduced number of scans. In this paper, the possibility of accurate image reconstruction with fewer scans is discussed. It is proved that the number of omitted scans cannot exceed the degree of the image for unique recovery. This necessary condition is one of the two major results of this paper. The second result is a new sampling method, syncopated sampling, which is proved to be effective even when quite few scans are performed. Syncopated sampling is compared to other suggested methods, demonstrating that this method recovers a unique image for a large variety of classes where other methods fail.

1. Introduction

MR imaging is used in various fields including medical imaging, chemistry, and physics. The imaging process works as follows: an MRI machine uses a powerful magnetic field to align magnetization, and radio frequency fields to systematically alter the alignment of this magnetization [3]. At each step, the MRI machine observes the radio frequencies through magnetic coils. After a sufficient number of scans, it processes the total data and constructs the image.

Although MR is advantageous in many ways, this process has the following drawbacks. First, MR imaging is a slow process that requires the object to stay still until the scan is complete. Therefore, this process has poor accuracy in monitoring objects in motion, e.g., a beating heart, blood vessels, or lungs. Secondly, it is a costly process, and thirdly, the map of coil sensitivities is hard to calculate.

The coil sensitivity issue is addressed by the inverse blind problem. The inverse blind problem (IBP) is to reconstruct the MR image when coil sensitivities are unknown. There are two major steps to solve this problem: the first is to show that the solution to IBP is unique, meaning that the data observed by the coils cannot suggest two different images. The second is to find algorithms that recover this unique solution from the observed data set.

When the total data is known, IBP has a unique solution and there are efficient algorithms to reconstruct the image from the data set [4]. However, solving IBP using less information is revolutionary since it reduces the cost of the process and increases the accuracy of imaging moving objects. One of the two major issues of this approach is whether it is always possible to reconstruct the MR image using less information; if the answer is positive, which sets of data suffice to reconstruct

the image? The second issue is finding reconstruction algorithms using that reduced data.

Recent studies have suggested recovery algorithms using a reduced amount of data [1, 2]. In this paper, it is proved that any accurate sampling method guaranteeing uniqueness cannot omit too many coefficients. A new method of data sampling, syncopated sampling, is presented. It is proved that under a mild condition on the image, this new method yields a unique solution to IBP, even when the set of observed data is very small.

In the Background section, the mathematical model of MR imaging is presented, uniform sampling is defined, and the assumptions on filters are established. In the Necessary Condition for Uniqueness section, the necessary condition for accurate samplings is presented and investigated alongside the method in [1]. In the Syncopated Sampling section, syncopated sampling is defined and its properties are discussed. In the Discussion section, results are summarized and open questions are posed for further investigation. In the Proofs section, the proofs of the theorems are presented.

2. Background

The mathematical setup in [7] is adopted: the image of the object is assumed to be a polynomial with complex coefficients of degree N, denoted by f:

$$f(x) = a_N x^N + a_{N-1} x^{N-1} + \cdots + a_1 x + a_0.$$

Let K denote the number of coils. Channels of the coils are assumed to be polynomials with complex coefficients of degrees $\delta_1, \delta_2, \ldots, \delta_K$, denoted by $h_1, h_2, \ldots, h_K$

respectively:

$$h_j = b_{j,\delta_j} x^{\delta_j} + b_{j,\delta_j - 1} x^{\delta_j - 1} + \cdots + b_{j,1} x + b_{j,0}.$$

In this setup, the radio frequencies observed through the coils at time i correspond to the coefficients of x^i in $fh_1, fh_2, \ldots, fh_K$, respectively.

The inverse blind problem imposes the condition that the coefficients of f and those of the h_j's are unknown, while certain coefficients of f multiplied by the h_j's are known. The aim is to recover f from this information. In this setup, this corresponds to finding the roots of f. If no conditions are imposed on the h_j's, then for almost any image f, one can choose the coefficients of h_j suitably so that fh_j yields the exact observed data. This means IBP is impossible to solve under no assumptions on the h_j's. Moreover, this does not align with physical conditions either; e.g., if the h_j share a common root, then the coils cannot distinguish between a large variety of different objects, which in reality is not the case. Therefore, we must make realistic assumptions on the filters. The first assumption is that the coils have adequate sensitivity to identify different objects, or in mathematical terminology:

$$h_1, h_2, \ldots, h_K \text{ share no common root.}$$

Under this assumption on h_j, and when all coefficients of $fh_1, fh_2, \ldots, fh_K$ along with the degree of f are known, f is uniquely recovered by the Euclidean algorithm [4]. In this case, all of the observed data is needed to recover the image. Our interest is to investigate whether one can find the roots of f with fewer coefficients.

Definition 2.1 (Uniform Sampling)**.** *Let r be a positive integer greater than* 1*. The data set consisting of the coefficients of each fh_j whose indices are congruent to* 0 (mod r) *is called uniform sampling.*

Here r stands for the frequency of lines observed. It is noted that in order to obtain accurate results using fewer coefficients, more restrictive conditions must be imposed on H. We adopt the condition in [4], since it holds for a very large variety of filter sets and helps reduce technical difficulties in proofs. For each $j = 1, 2, \ldots, K$, h_j can be uniquely written as

$$h_j(x) = h_j^{(0)}(x^r) + xh_j^{(1)}(x^r) + x^2h_j^{(2)}(x^r) + \cdots + x^{r-1}h_j^{(r-1)}(x^r). \tag{1}$$

Let H be

$$H = \begin{pmatrix} h_1^{(0)} & h_1^{(r-1)} & h_1^{(r-2)} & \ldots & h_1^{(1)} \\ h_2^{(0)} & h_2^{(r-1)} & h_2^{(r-2)} & \ldots & h_2^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h_K^{(0)} & h_K^{(r-1)} & h_K^{(r-2)} & \ldots & h_K^{(1)} \end{pmatrix}.$$

Definition 2.2. *H is called a totally unimodular polynomial matrix if every square $r \times r$ submatrix of H is unimodular.*

The final assumption on the filters is that H is totally unimodular. All assumptions made up to this point are that the degree of f, the degrees of the h_j's, and certain coefficients of the fh_j's are known, and that H is totally unimodular.

3. Necessary Condition for Uniqueness

In this section, the necessary condition is presented.

Theorem 3.1. *Assume that N is less than or equal to the number of unobserved coefficients, and also assume that the h_j's are given and fixed. Then, independent of which coefficients are observed, the set*

$$\big\{(\alpha_1, \alpha_2, \ldots, \alpha_N) \in \mathbb{C}^N \;\big|\; \exists c \in \mathbb{C} : f(x) = c(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_N), \text{ and } fh_j \text{ gives the exact same observed output for } j = 1, 2, \ldots, K\big\}$$

is open and dense in $\mathbb{C}^N$ in the usual metric topology.

This theorem states that, regardless of what condition is imposed on the h_j's, the number of unobserved coefficients cannot exceed the degree of f. In other words, if the unobserved coefficients exceed the degree of the image, the MRI machine will fail to identify different types of objects.

In [1], it was shown that uniform sampling plus $r-1$ consecutive coefficients of fh_j yields accurate results in solving IBP. However, applying Theorem 3.1 to this case yields the following result. There are $\lceil \frac{N+\delta}{r} \rceil$ observed coefficients from uniform sampling and $r-1$ observed coefficients from consecutive coefficients. Hence, the number of unobserved coefficients is $N + \delta - \lceil \frac{N+\delta}{r} \rceil - (r-1)$, where δ is the largest of the δ_j's. If this quantity is greater than or equal to N, the theorem states that unique recovery is impossible. That is:

Remark 3.2. *If*

$$N \leq N + \delta - \left\lceil \frac{N+\delta}{r} \right\rceil - (r-1),$$

i.e.,

$$N \leq (\delta - r - 1)(r-1),$$

then unique recovery for the IBP is not possible.

This remark shows that the method in [1] may fail to recover a unique image in a variety of cases. This result seems contradictory to the accurate experimental results in [1, 2]. The explanation is that current medical imaging imposes constraints on r, N, δ, and K. Reasonable values for these quantities are:

$$2 \leq r \leq 10, \quad 5 \leq \delta \leq 23, \quad 128 \leq N \leq 1024, \quad 8 \leq K \leq 32.$$

Under these constraints, one can easily check that the inequality in Remark 3.2 is not satisfied, and thus the non-uniqueness result is not applicable. Although requiring a formal mathematical proof, experiments suggest that when the observed data meets the threshold implied by Remark 3.2, the image can be uniquely recovered for a large variety of classes. This explains why the experimental results in [1] are accurate. On the other hand, under more advanced physical conditions—e.g., an increased number of coils, or when r exceeds 10—that sampling method fails to guarantee unique recovery of the image.

4. Syncopated Sampling

In this section, a new sampling method, syncopated sampling, is presented. It is shown that this method works accurately for unique recovery across a larger variety of cases where the method in [1] may fail.

Definition 4.1 (Syncopated Sampling)**.** *The data set consisting of the coefficients of each fh_j whose indices are congruent to* 0 *or* $1 \pmod{r}$ *is called syncopated sampling.*

Theorem 4.2 (Uniqueness). *Let $\delta_{s_1} \leq \delta_{s_2} \leq \cdots \leq \delta_{s_K}$ be the ordering of the degrees of the channels, where $\{\delta_{s_1}, \delta_{s_2}, \ldots, \delta_{s_K}\} = \{\delta_1, \delta_2, \ldots, \delta_K\}$, and let $m = 2\lfloor \delta_{s_r} \rfloor + 2$. Assume that*

$$
\begin{aligned}
&f^{(0)}, zf^{(0)}, z^2 f^{(0)}, \ldots, z^m f^{(0)}, \\
&f^{(1)}, zf^{(1)}, z^2 f^{(1)}, \ldots, z^m f^{(1)}, \\
&\vdots \\
&f^{(r-1)}, zf^{(r-1)}, z^2 f^{(r-1)}, \ldots, z^m f^{(r-1)}
\end{aligned}
$$

are linearly independent over $\mathbb{C}$. Then $H' = cI_{r\times r}$ for some $c \in \mathbb{C}$, i.e., f is unique up to a constant multiple.

This theorem shows that syncopated sampling yields a unique solution to IBP when the linear independence condition is satisfied. We note that this condition is not overly restrictive. The following example shows that syncopated sampling guarantees uniqueness in cases where the method in [1] fails.

Consider three different classes of objects whose images are denoted by f_1, f_2, and f_3. Take $N = 128$, $\delta_i = 23$, $h_i = 1 + x^i + x^{23}$, and $r = 11$. Set f_1, f_2, and f_3 to be:

$$
\begin{aligned}
f_1 &= x^{128} + 1, \\
f_2 &= x^{128} + bx + 1, \\
f_3 &= x^{128} + dx^{11T+k} - dx^{11T+24-k} + 1,
\end{aligned}
$$

where b, d are arbitrary non-zero complex numbers with $b \neq d$, $T \leq 11$ is a positive integer, and $k \not\equiv 0, 2, 4, 9 \pmod{11}$ is a positive integer not exceeding 24. Note that

the h_j's satisfy the unimodularity condition and f_1, f_2, f_3 satisfy the condition in Theorem 4.2. Under uniform sampling plus $r-1$ consecutive coefficients, f_1h_j, f_2h_j, and f_3h_j read their

$$\{0, 11, 22, 33, 44, 55, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 88, 99, 110, 121, 132, 143\}$$

-th coefficients identically as:

$$\{1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\}.$$

In syncopated sampling, however, the 12-th coefficients of f_1h_1, f_2h_1, and f_3h_k are 0, b, and d respectively, which are mutually distinct. Therefore, syncopated sampling distinguishes between f_1, f_2, and f_3.

First, remark that there are 7 possibilities for k and 11 for T; hence, there are at least 77 distinct cases that produce identical data under the method in [1]. Second, by increasing r and K, one can construct numerous similar examples that are impossible to distinguish under the method in [1]. Third, it is natural to ask whether syncopated sampling fails to produce a unique image when it has too few coefficients and violates the condition of the uniqueness theorem. On the contrary, syncopated sampling remains accurate due to the linear independence condition on the image. Although this condition is mild, it suffices to guarantee uniqueness.

5. Discussion

In this paper, we showed that any accurate algorithm solving IBP requires a number of coefficients at least equal to the total degree minus the degree of the image. It was demonstrated that the method in [1] may fail to produce a unique image.

A new method, syncopated sampling, was presented and its advantages over the previous sampling method were discussed. It is conjectured that the uniqueness theorem is an "if and only if" statement—that is, whenever the number of observed coefficients is sufficient in terms of the uniqueness theorem, one can recover the image uniquely. This statement requires a formal mathematical proof. Furthermore, no explicit recovery algorithm was presented for syncopated sampling in this paper; further investigation is needed to develop reconstruction algorithms for syncopated sampling.

6. Proofs

Proof of Theorem 3.1. We have that the number of unobserved coefficients is at least N. Let $k_1, k_2, \ldots, k_N$ be N unobserved coefficient indices. Let $(\alpha_1, \alpha_2, \ldots, \alpha_N) \in \mathbb{C}^N$ be such that

$$\begin{pmatrix} \alpha_1^{k_1} & \alpha_1^{k_2} & \ldots & \alpha_1^{k_N} \\ \alpha_2^{k_1} & \alpha_2^{k_2} & \ldots & \alpha_2^{k_N} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_N^{k_1} & \alpha_N^{k_2} & \ldots & \alpha_N^{k_N} \end{pmatrix}$$

is invertible. It is known that such a set is open and dense in $\mathbb{C}^N$ in the metric topology. Now we prove that we can fill the unobserved coefficients such that the observed coefficients of fh_j's, where $f(x) = C(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_N)$, coincide with the original output. Set every unobserved coefficient of fh_j, except $k_1, k_2, \ldots, k_N$, to be 0 for all $j = 1, 2, \ldots, K$. We require α_i to be a root of fh_j for every $i = 1, 2, \ldots, N$ and $j = 1, 2, \ldots, K$. Writing this out:

$$(2)\quad 0 = f(\alpha_i)h_j(\alpha_i) = A_{k_1}^{(j)}\alpha_i^{k_1} + A_{k_2}^{(j)}\alpha_i^{k_2} + \cdots + A_{k_N}^{(j)}\alpha_i^{k_N} + B_{o_1}^{(j)}\alpha_i^{o_1} + B_{o_2}^{(j)}\alpha_i^{o_2} + \cdots + B_{o_L}^{(j)}\alpha_i^{o_L},$$

where $A^{(j)}_{k_1}, \ldots, A^{(j)}_{k_N}$ and $B^{(j)}_{o_1}, \ldots, B^{(j)}_{o_L}$ denote the unobserved and observed coefficients of fh_j, respectively. Set

$$u_j(\alpha_i) = B^{(j)}_{o_1}\alpha_i^{o_1} + B^{(j)}_{o_2}\alpha_i^{o_2} + \cdots + B^{(j)}_{o_L}\alpha_i^{o_L}.$$

Then (2) becomes:

$$A^{(j)}_{k_1}\alpha_i^{k_1} + A^{(j)}_{k_2}\alpha_i^{k_2} + \cdots + A^{(j)}_{k_N}\alpha_i^{k_N} = -u_j(\alpha_i).$$

Writing this for $i = 1, 2, \ldots, N$ and $j = 1, 2, \ldots, K$, we obtain the matrix equation:

(3)
$$\begin{pmatrix} A^{(1)}_{k_1} & A^{(1)}_{k_2} & \ldots & A^{(1)}_{k_N} \\ A^{(2)}_{k_1} & A^{(2)}_{k_2} & \ldots & A^{(2)}_{k_N} \\ \vdots & \vdots & \ddots & \vdots \\ A^{(K)}_{k_1} & A^{(K)}_{k_2} & \ldots & A^{(K)}_{k_N} \end{pmatrix} \begin{pmatrix} \alpha_1^{k_1} & \alpha_2^{k_1} & \ldots & \alpha_N^{k_1} \\ \alpha_1^{k_2} & \alpha_2^{k_2} & \ldots & \alpha_N^{k_2} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k_N} & \alpha_2^{k_N} & \ldots & \alpha_N^{k_N} \end{pmatrix} = \begin{pmatrix} -u_1(\alpha_1) & -u_1(\alpha_2) & \ldots & -u_1(\alpha_N) \\ -u_2(\alpha_1) & -u_2(\alpha_2) & \ldots & -u_2(\alpha_N) \\ \vdots & \vdots & \ddots & \vdots \\ -u_K(\alpha_1) & -u_K(\alpha_2) & \ldots & -u_K(\alpha_N) \end{pmatrix}.$$

Multiplying both sides of (3) on the right by

$$\begin{pmatrix} \alpha_1^{k_1} & \alpha_2^{k_1} & \ldots & \alpha_N^{k_1} \\ \alpha_1^{k_2} & \alpha_2^{k_2} & \ldots & \alpha_N^{k_2} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k_N} & \alpha_2^{k_N} & \ldots & \alpha_N^{k_N} \end{pmatrix}^{-1},$$

we obtain a set of values for the unobserved (and yet unfilled) coefficients, which guarantee that $f(\alpha_i)h_j(\alpha_i) = 0$. Choosing the α_i's distinct from the roots of h_j (a finite set of roots), we guarantee that $f(\alpha_i) = 0$ for all $i = 1, 2, \ldots, N$. Since f has degree N,

$$f(x) = C(x - \alpha_1)(x - \alpha_2)\ldots(x - \alpha_N)$$

for some non-zero $C \in \mathbb{C}$. □

Proof of Theorem 4.2. Suppose that for another channel system–image pair, we obtain the same syncopated sampling. Denote the new channels by $\{h'_1, h'_2, \ldots, h'_K\}$ and the new image by f'. It is convenient to make the following two assumptions: first, that the polynomial matrix

$$H' = \begin{pmatrix} h'^{(0)}_1 & h'^{(r-1)}_1 & \ldots & h'^{(1)}_1 \\ h'^{(0)}_2 & h'^{(r-1)}_2 & \ldots & h'^{(1)}_2 \\ \vdots & \vdots & \ddots & \vdots \\ h'^{(0)}_K & h'^{(r-1)}_K & \ldots & h'^{(1)}_K \end{pmatrix}$$

is a totally unimodular polynomial matrix, and second, that the degrees of the channel system do not change, i.e., $\deg(h_j) = \deg(h'_j)$ for all $j = 1, 2, \ldots, K$. Define F' to be

$$F' = \begin{pmatrix} f'^{(0)} & f'^{(1)} \\ zf'^{(1)} & zf'^{(2)} \\ zf'^{(2)} & zf'^{(3)} \\ \vdots & \vdots \\ zf'^{(r-2)} & zf'^{(r-1)} \\ zf'^{(r-1)} & f'^{(0)} \end{pmatrix}.$$

We have that

$$Y = HF = H'F'.$$

Let H_r, H'_r, and Y_r denote the $r \times r$ matrices composed of rows $l_1, l_2, \ldots, l_r$ of H, H', and Y respectively, where $\{l_1, l_2, \ldots, l_r\} \subset \{1, 2, \ldots, K\}$ (we will choose suitable $l_1, l_2, \ldots, l_r$ below). The previous equation implies that

$$Y_r = H_r F = H'_r F'.$$

Since H' is totally unimodular, H'_r is invertible. Hence,

$$F' = (H'_r)^{-1} H_r F. \tag{4}$$

Let us write $(H'_r)^{-1}H_r$ as

$$(H'_r)^{-1}H_r = \begin{pmatrix} h'_{1,1} & h'_{1,2} & \dots & h'_{1,r} \\ h'_{2,1} & h'_{2,2} & \dots & h'_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ h'_{r,1} & h'_{r,2} & \dots & h'_{r,r} \end{pmatrix},$$

where $h'_{i,j} \in \mathbb{C}[z]$. Rewriting (4):

$$\begin{pmatrix} h'_{1,1} & h'_{1,2} & \dots & h'_{1,r} \\ h'_{2,1} & h'_{2,2} & \dots & h'_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ h'_{r,1} & h'_{r,2} & \dots & h'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(0)} & f^{(1)} \\ zf^{(1)} & zf^{(2)} \\ zf^{(2)} & zf^{(3)} \\ \vdots & \vdots \\ zf^{(r-2)} & zf^{(r-1)} \\ zf^{(r-1)} & f^{(0)} \end{pmatrix} = \begin{pmatrix} f'^{(0)} & f'^{(1)} \\ zf'^{(1)} & zf'^{(2)} \\ zf'^{(2)} & zf'^{(3)} \\ \vdots & \vdots \\ zf'^{(r-2)} & zf'^{(r-1)} \\ zf'^{(r-1)} & f'^{(0)} \end{pmatrix}.$$

That is,

$$\begin{pmatrix} h'_{1,1} & h'_{1,2} & \dots & h'_{1,r} \\ h'_{2,1} & h'_{2,2} & \dots & h'_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ h'_{r,1} & h'_{r,2} & \dots & h'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(0)} \\ zf^{(1)} \\ zf^{(2)} \\ \vdots \\ zf^{(r-1)} \end{pmatrix} = \begin{pmatrix} f'^{(0)} \\ zf'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-1)} \end{pmatrix} \tag{5}$$

and

$$\begin{pmatrix} h'_{1,1} & h'_{1,2} & \dots & h'_{1,r} \\ h'_{2,1} & h'_{2,2} & \dots & h'_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ h'_{r,1} & h'_{r,2} & \dots & h'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(1)} \\ zf^{(2)} \\ \vdots \\ zf^{(r-1)} \\ f^{(0)} \end{pmatrix} = \begin{pmatrix} f'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-1)} \\ f'^{(0)} \end{pmatrix}. \tag{6}$$

By modifying the LHS of (5), one obtains:

$$\begin{pmatrix} h'_{1,1} & zh'_{1,2} & zh'_{1,3} & \dots & zh'_{1,r-1} & zh'_{1,r} \\ h'_{2,1} & zh'_{2,2} & zh'_{2,3} & \dots & zh'_{2,r-1} & zh'_{2,r} \\ h'_{3,1} & zh'_{3,2} & zh'_{3,3} & \dots & zh'_{3,r-1} & zh'_{3,r} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h'_{r,1} & zh'_{r,2} & zh'_{r,3} & \dots & zh'_{r,r-1} & zh'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(0)} \\ f^{(1)} \\ f^{(2)} \\ \vdots \\ f^{(r-2)} \\ f^{(r-1)} \end{pmatrix} = \begin{pmatrix} f'^{(0)} \\ f'^{(1)} \\ f'^{(2)} \\ \vdots \\ f'^{(r-2)} \\ f'^{(r-1)} \end{pmatrix}. \tag{7}$$

Similarly, by modifying the LHS of (6), one obtains:

$$\begin{pmatrix} h'_{1,1} & zh'_{1,2} & zh'_{1,3} & \dots & zh'_{1,r-1} & h'_{1,r} \\ h'_{2,1} & zh'_{2,2} & zh'_{2,3} & \dots & zh'_{2,r-1} & h'_{2,r} \\ h'_{3,1} & zh'_{3,2} & zh'_{3,3} & \dots & zh'_{3,r-1} & h'_{3,r} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h'_{r,1} & zh'_{r,2} & zh'_{r,3} & \dots & zh'_{r,r-1} & h'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(1)} \\ f^{(2)} \\ \vdots \\ f^{(r-2)} \\ f^{(r-1)} \\ f^{(0)} \end{pmatrix} = \begin{pmatrix} f'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-2)} \\ zf'^{(r-1)} \\ f'^{(0)} \end{pmatrix}. \tag{8}$$

Multiplying both sides of (8) on the left by

$$\begin{pmatrix} 0 & 0 & \dots & 0 & 1 \\ z & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix},$$

one gets:

$$\begin{pmatrix} 0 & 0 & \dots & 0 & 1 \\ z & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix} \begin{pmatrix} h'_{1,1} & zh'_{1,2} & zh'_{1,3} & \dots & zh'_{1,r-1} & h'_{1,r} \\ h'_{2,1} & zh'_{2,2} & zh'_{2,3} & \dots & zh'_{2,r-1} & h'_{2,r} \\ h'_{3,1} & zh'_{3,2} & zh'_{3,3} & \dots & zh'_{3,r-1} & h'_{3,r} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h'_{r,1} & zh'_{r,2} & zh'_{r,3} & \dots & zh'_{r,r-1} & h'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(1)} \\ f^{(2)} \\ \vdots \\ f^{(r-2)} \\ f^{(r-1)} \\ f^{(0)} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & \dots & 0 & 1 \\ z & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix} \begin{pmatrix} f'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-2)} \\ zf'^{(r-1)} \\ f'^{(0)} \end{pmatrix} = \begin{pmatrix} f'^{(0)} \\ zf'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-1)} \end{pmatrix}.$$

Substituting the identity

$$\begin{pmatrix} f^{(1)} \\ f^{(2)} \\ \vdots \\ f^{(r-2)} \\ f^{(r-1)} \\ f^{(0)} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} f^{(0)} \\ f^{(1)} \\ \vdots \\ f^{(r-3)} \\ f^{(r-2)} \\ f^{(r-1)} \end{pmatrix}$$

into the previous equation gives:

$$\begin{pmatrix} 0 & 0 & \dots & 0 & 1 \\ z & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix} \begin{pmatrix} h'_{1,1} & zh'_{1,2} & zh'_{1,3} & \dots & zh'_{1,r-1} & h'_{1,r} \\ h'_{2,1} & zh'_{2,2} & zh'_{2,3} & \dots & zh'_{2,r-1} & h'_{2,r} \\ h'_{3,1} & zh'_{3,2} & zh'_{3,3} & \dots & zh'_{3,r-1} & h'_{3,r} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h'_{r,1} & zh'_{r,2} & zh'_{r,3} & \dots & zh'_{r,r-1} & h'_{r,r} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} f^{(0)} \\ f^{(1)} \\ \vdots \\ f^{(r-3)} \\ f^{(r-2)} \\ f^{(r-1)} \end{pmatrix}$$

$$= \begin{pmatrix} f'^{(0)} \\ zf'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-1)} \end{pmatrix},$$

which simplifies to

$$\begin{pmatrix} h'_{r,r} & h'_{r,1} & zh'_{r,2} & zh'_{r,3} & \dots & zh'_{r,r-1} \\ zh'_{1,r} & zh'_{1,1} & z^2h'_{1,2} & z^2h'_{1,3} & \dots & z^2h'_{1,r-1} \\ h'_{2,r} & h'_{2,1} & zh'_{2,2} & zh'_{2,3} & \dots & zh'_{2,r-1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ h'_{r-1,r} & h'_{r-1,1} & zh'_{r-1,2} & zh'_{r-1,3} & \dots & h'_{r-1,r-1} \end{pmatrix} \begin{pmatrix} f^{(0)} \\ f^{(1)} \\ \vdots \\ f^{(r-3)} \\ f^{(r-2)} \\ f^{(r-1)} \end{pmatrix} = \begin{pmatrix} f'^{(0)} \\ zf'^{(1)} \\ zf'^{(2)} \\ \vdots \\ zf'^{(r-1)} \end{pmatrix}.$$

Subtracting this result from (7), one obtains:

(9)

$$\begin{pmatrix} h'_{r,r} - h'_{1,1} & h'_{r,1} - zh'_{1,2} & zh'_{r,2} - zh'_{1,3} & zh'_{r,3} - zh'_{1,4} & \dots & zh'_{r,r-1} - zh'_{1,r} \\ zh'_{1,r} - h'_{2,1} & zh'_{1,1} - zh'_{2,2} & z^2h'_{1,2} - zh'_{2,3} & z^2h'_{1,3} - zh'_{2,4} & \dots & z^2h'_{1,r-1} - zh'_{2,r} \\ h'_{2,r} - h'_{3,1} & h'_{2,1} - zh'_{3,2} & zh'_{2,2} - zh'_{3,3} & zh'_{2,3} - zh'_{3,4} & \dots & zh'_{2,r-1} - zh'_{3,r} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ h'_{r-1,r} - h'_{r,1} & h'_{r-1,1} - zh'_{r,2} & zh'_{r-1,2} - zh'_{r,3} & zh'_{r-1,3} - zh'_{r,4} & \dots & zh'_{r-1,r-1} - zh'_{r,r} \end{pmatrix} \begin{pmatrix} f^{(0)} \\ f^{(1)} \\ \vdots \\ f^{(r-3)} \\ f^{(r-2)} \\ f^{(r-1)} \end{pmatrix} = 0.$$

Choose $\{l_1, l_2, \dots, l_r\}$ such that $\{l_1, l_2, \dots, l_r\} = \{s_1, s_2, \dots, s_r\}$. Then the degrees of H'_r and H_r are at most $\lfloor \delta_{s_r} \rfloor$. By the previous result on the degree of the inverse of unimodular polynomial matrices, $(H'_r)^{-1}$ has degree at most $\lfloor \delta_{s_r} \rfloor$. Therefore, $(H'_r)^{-1}H_r$ has degree at most $2\lfloor \delta_{s_r} \rfloor$, i.e., $\deg(h'_{i,j}) \leq 2\lfloor \delta_{s_r} \rfloor$ for all $(i,j) \in \{1,2,\dots,r\} \times \{1,2,\dots,r\}$. Let us denote the z^k-th coefficient of a polynomial $g \in \mathbb{C}[z]$ by $g_{(k)}$. Writing the first row of (9), one gets:

$$\begin{aligned} 0 = &\left(h'_{r,r} - h'_{1,1}\right) f^{(0)} + \left(h'_{r,1} - zh'_{1,2}\right) f^{(1)} + \left(zh'_{r,2} - zh'_{1,3}\right) f^{(2)} \\ &+ \left(zh'_{r,3} - zh'_{1,4}\right) f^{(3)} + \dots + \left(zh'_{r,r-1} - zh'_{1,r}\right) f^{(r-1)}. \end{aligned}$$

Expanding into powers of z,

$$0 = \sum_{k_1} \left(h'_{r,r} - h'_{1,1}\right)_{(k_1)} z^{k_1} f^{(0)} + \sum_{k_2} \left(h'_{r,1} - zh'_{1,2}\right)_{(k_2)} z^{k_2} f^{(1)} + \sum_{k_3} \left(zh'_{r,2} - zh'_{1,3}\right)_{(k_3)} z^{k_3} f^{(2)}$$
$$+ \sum_{k_4} \left(zh'_{r,3} - zh'_{1,4}\right)_{(k_4)} z^{k_4} f^{(3)} + \cdots + \sum_{k_r} \left(zh'_{r,r-1} - zh'_{1,r}\right)_{(k_r)} z^{k_r} f^{(r-1)},$$

which by linear independence implies that

$$0 = \left(h'_{r,r} - h'_{1,1}\right)_{(k_1)} = \left(h'_{r,1} - zh'_{1,2}\right)_{(k_2)} = \left(zh'_{r,2} - zh'_{1,3}\right)_{(k_3)} = \cdots = \left(zh'_{r,r-1} - zh'_{1,r}\right)_{(k_r)}$$

for all $k_1, k_2, \ldots, k_r \leq m$, since $\deg(h'_{i,j}) \leq 2\lfloor \delta_{s_r} \rfloor$ implies $k_1, k_2, \ldots, k_r \leq m$. This is equivalent to

$$0 = h'_{r,r} - h'_{1,1} = h'_{r,1} - zh'_{1,2} = zh'_{r,2} - zh'_{1,3} = zh'_{r,3} - zh'_{1,4} = \cdots = zh'_{r,r-1} - zh'_{1,r}. \tag{10}$$

Applying a similar process to all remaining rows, we obtain:

$$\begin{pmatrix} h'_{r,r} - h'_{1,1} & h'_{r,1} - zh'_{1,2} & zh'_{r,2} - zh'_{1,3} & zh'_{r,3} - zh'_{1,4} & \cdots & zh'_{r,r-1} - zh'_{1,r} \\ zh'_{1,r} - h'_{2,1} & zh'_{1,1} - zh'_{2,2} & z^2h'_{1,2} - zh'_{2,3} & z^2h'_{1,3} - zh'_{2,4} & \cdots & z^2h'_{1,r-1} - zh'_{2,r} \\ h'_{2,r} - h'_{3,1} & h'_{2,1} - zh'_{3,2} & zh'_{2,2} - zh'_{3,3} & zh'_{2,3} - zh'_{3,4} & \cdots & zh'_{2,r-1} - zh'_{3,r} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ h'_{r-1,r} - h'_{r,1} & h'_{r-1,1} - zh'_{r,2} & zh'_{r-1,2} - zh'_{r,3} & zh'_{r-1,3} - zh'_{r,4} & \cdots & zh'_{r-1,r-1} - zh'_{r,r} \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

From the diagonal entries of the matrix above, we get

$$h'_{1,1} = h'_{2,2} = h'_{3,3} = \cdots = h'_{r,r} = a_1, \tag{11}$$

where $a_1 \in \mathbb{C}[z]$. From the (i,j)-th entries where $i - j \equiv 1 \pmod{r}$, we get

$$zh'_{1,2} = h'_{2,3} = \cdots = h'_{r-1,r} = h'_{r,1} = za_2, \tag{12}$$

where $a_2 \in \mathbb{C}[z]$. From the (i,j)-th entries where $i - j \equiv 2 \pmod{r}$, we get

$$zh'_{1,3} = h'_{2,4} = \cdots = h'_{r-2,r} = h'_{r-1,1} = zh'_{r,2} = za_3, \tag{13}$$

where $a_3 \in \mathbb{C}[z]$. From the (i,j)-th entries where $i - j \equiv 3 \pmod{r}$, we get

$$zh'_{1,4} = h'_{2,5} = \cdots = h'_{r-3,r} = h'_{r-2,2} = zh'_{r-1,2} = zh'_{r,3} = za_4, \tag{14}$$

where $a_4 \in \mathbb{C}[z]$. Similarly, from the (i,j)-th entries where $i - j \equiv k \pmod{r}$, we get

$$zh'_{1,k+1} = h'_{2,k+2} = \cdots = h'_{r-k,r} = h'_{r-k+1,1} = zh'_{r-k+2,2} = zh'_{r-k+3,3} = \cdots = zh'_{r,k} = za_k, \tag{15}$$

where $a_k \in \mathbb{C}[z]$. Combining these results, $(H'_r)^{-1}H_r$ has the form:

$$(H'_r)^{-1}H_r = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 & a_5 & \dots & a_{r-1} & a_r \\ za_r & a_1 & za_2 & za_3 & za_4 & \dots & za_{r-2} & za_{r-1} \\ za_{r-1} & a_r & a_1 & za_2 & za_3 & \dots & za_{r-3} & za_{r-2} \\ za_{r-2} & a_{r-1} & a_r & a_1 & za_2 & \dots & za_{r-4} & za_{r-3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ za_3 & a_4 & a_5 & a_6 & a_7 & \dots & a_1 & za_2 \\ za_2 & a_3 & a_4 & a_5 & a_6 & \dots & a_r & a_1 \end{pmatrix} \tag{16}$$

for some $a_1, a_2, \dots, a_r \in \mathbb{C}[z]$.

We know that $(H'_r)^{-1}H_r$ is a unimodular polynomial matrix, i.e., $\det((H'_r)^{-1}H_r) = c$ for some $c \in \mathbb{C}$. On the other hand, expanding $\det((H'_r)^{-1}H_r) = c$ from (16), the coefficient of

z^{r-2} is a_2^r, hence $a_2 = 0$. Substituting $a_2 = 0$ into (16) yields:

$$(H_r')^{-1}H_r = \begin{pmatrix} a_1 & 0 & a_3 & a_4 & a_5 & \dots & a_{r-1} & a_r \\ za_r & a_1 & 0 & za_3 & za_4 & \dots & za_{r-2} & za_{r-1} \\ za_{r-1} & a_r & a_1 & 0 & za_3 & \dots & za_{r-3} & za_{r-2} \\ za_{r-2} & a_{r-1} & a_r & a_1 & 0 & \dots & za_{r-4} & za_{r-3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ za_3 & a_4 & a_5 & a_6 & a_7 & \dots & a_1 & 0 \\ 0 & a_3 & a_4 & a_5 & a_6 & \dots & a_r & a_1 \end{pmatrix}. \tag{17}$$

Similarly, the coefficient of z^{r-3} in (17) is a_3^r, hence $a_3 = 0$. Substituting $a_3 = 0$ into (17) and applying this procedure inductively, after $r-1$ steps one obtains:

$$(H_r')^{-1}H_r = a_1 I_{r\times r}. \tag{18}$$

Unimodularity of $(H_r')^{-1}H_r$ implies that $a_1 \in \mathbb{C}$. Hence, $F' = a_1 F$, i.e., $f' = a_1 f$, so f is unique up to a constant multiple. The proof is complete. □

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